DIGITAL FORENSICS · IMAGE EXAMINATION

Image Forensic Examination Report

Evidence file: **edited_metadata.jpg**

CASE ID	EXAMINER	GENERATED (UTC)	TOOL
PP-20261103-174844-2509	Sagar	2026-11-03 17:48:44	PixelProof 1.0.0

Suspicious

At least one signal warrants examiner review.

Confidence 90% · coverage 3 module(s) · agreement 70%

20.0
out of 100

Banding: score 20 to 49 · final score blends the weighted consensus (10.9) with the strongest single module (30.0).

1. Executive summary

PixelProof executed **3** analysis module(s) in **1.2 s**. The combined manipulation score is **20.0/100**, placing the image in the **Suspicious** band (score 20 to 49). The strongest contribution came from **Metadata / EXIF** (30.0/100). **2** finding(s) above info level were recorded: 0 high, 1 medium, 1 low.

Signals supporting manipulation

- Metadata / EXIF scored 30.0/100 - 2 EXIF tags · editor: Adobe Photoshop

Signals against manipulation

- File analysis & integrity scored 0.0/100 - JPEG · 1200×800 · 46.8 KB · signature/extension match
- Error Level Analysis scored 0.0/100 - mean residual 0.21 · p99.5 3.0 · 0 dense region(s) · outlier fraction 12.1%

At least one module crossed the review threshold, so the blended score was raised to the review band instead of reporting the image as likely authentic.
Not available in this build: Copy-move detection, Steganography checks. These modules are excluded from the score rather than scored as zero.

2. Evidence and integrity

FILE NAME	edited_metadata.jpg
FULL PATH	`samples/tampered/edited_metadata.jpg`
SIZE	46.8 KB (47,929 bytes)
CONTAINER	JPEG · magic JPEG · extension .jpg (match)
DIMENSIONS	1200 x 800 px · 0.96 MP · RGB mode
FRAMES	1
FILESYSTEM MODIFIED	2026-11-03T12:12:49.048492+00:00
ANALYSIS WINDOW	2026-11-03T17:48:43.372557+00:00 to 2026-11-03T17:48:44.547592+00:00 (1.18 s)
SHA-256	`25096558ef5d2338189c2c594fd0747b87a0c10a71aeb58716761878c06e7fc4`
SHA-1	`78c38ebba1e6989bf94a8abf20d1744e60c32f77`
MD5	`4a3aeabd646f06c277b9429dd92b313e`

Hashes cover the exact bytes on disk. Re-hashing the same file later must reproduce SHA-256 exactly; any difference invalidates this report's link to the evidence.

3. Module results

Module	Status	Weight	Signal	Score	Summary
File analysis & integrity	OK	0.27		0.0	JPEG · 1200×800 · 46.8 KB · signature/extension match
Metadata / EXIF	OK	0.36		30.0	2 EXIF tags · editor: Adobe Photoshop
Error Level Analysis	OK	0.36		0.0	mean residual 0.21 · p99.5 3.0 · 0 dense region(s) · outlier fraction 12.1%

Weights are re-normalised over the modules that ran, so a partial pipeline still produces a meaningful score. The Signal column is each module's own 0-100 manipulation signal, not the final verdict.

4. Findings

MEDIUM

Editing or processing software recorded

Metadata names Adobe Photoshop. Post-processing is not automatically malicious, but it breaks the camera-original chain and must be explained.

Source: Metadata / EXIF · Confidence 85% · editor, provenance

LOW

No camera make/model tags

EXIF exists but does not identify a camera. This happens after editing, social-media processing, screenshots and AI generation.

Source: Metadata / EXIF · Confidence 60% · metadata, provenance

INFO

Cryptographic digests computed

SHA-256 25096558ef5d2338189c2c594fd0747b87a0c10a71aeb58716761878c06e7fc4 · SHA-1 78c38ebba1e6989bf94a8abf20d1744e60c32f77 · MD5 4a3aeabd646f06c277b9429dd92b313e

Source: File analysis & integrity · Confidence 99% · integrity, custody

INFO

Filename and signature are consistent

JPEG content, 1200×800, extension '.jpg' accepted.

Source: File analysis & integrity · Confidence 95% · container

INFO

No compact ELA anomaly

12.1% of blocks are above the residual threshold, but they do not form a dense region larger than the minimum cluster size. Edges, texture and prior compression explain such scattered responses. This supports, but does not prove, a single compression history.

Source: Error Level Analysis · Confidence 60% · ela

5. Visual evidence

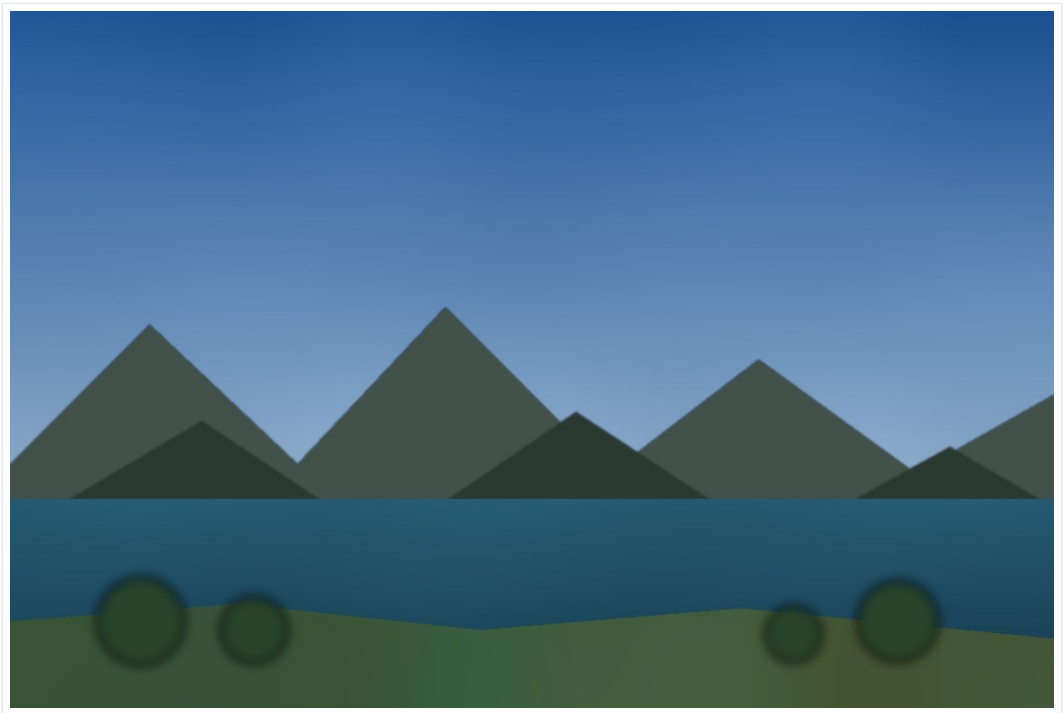


Figure 1 — Evidence image as submitted (display copy only).

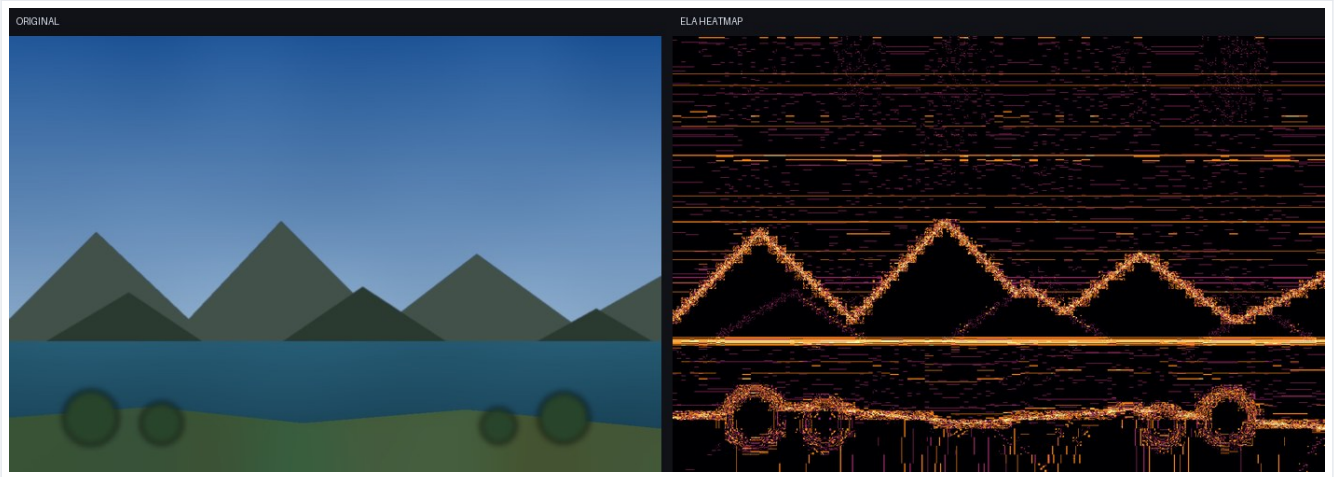


Figure 2 — Original (left) and ELA heatmap (right). Bright areas changed most after recompression at the configured quality.

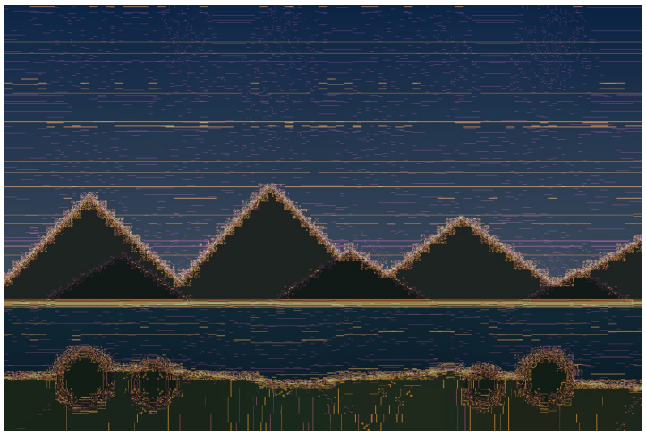


Figure 3 — ELA overlay with detected regions outlined

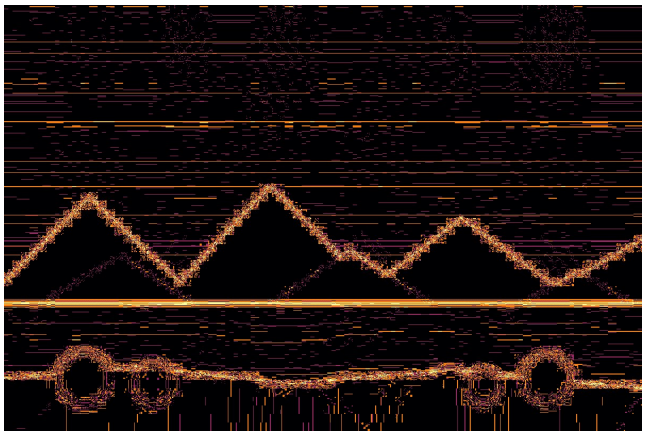


Figure 4 — Colourised ELA residual

6. Metadata detail

Camera and software

MAKE	—
MODEL	—
LENS	—
BODY SERIAL	—
SOFTWARE	Adobe Photoshop 25.0 (Windows)
ARTIST	—
COPYRIGHT	—

Location

No GPS block was present.

Timestamp coherence

Tag change time	2026-09-10 18:05:00 UTC
File modified	2026-11-03 12:12:49 UTC

Provenance keywords

Editing software	Adobe Photoshop
AI generators	none detected
EXIF tags read	2
ICC profile	absent

EXIF tags

Tag	Value
Image DateTime	2026:09:10 18:05:00
Image Software	Adobe Photoshop 25.0 (Windows)

7. Method and thresholds

Every threshold used below is stored in `utils/config.py`. They are empirical values chosen for this project's sample set, not trained on a labelled corpus. The report records them so another examiner can reproduce the run.

Module	Settings used
File analysis	MD5, SHA-1 and SHA-256 over the exact bytes; magic-byte container detection; extension agreement; double-extension regex; 50 MB and 80,000,000 pixel limits.
Metadata	ExifRead tag extraction; GPS rationals to decimal, DMS and OSM link; timestamp coherence checks; editing and AI keyword lists; EXIF thumbnail dHash compared with the main image (Hamming distance > 16 is a mismatch).
ELA	JPEG re-save at quality 90; 16 px blocks; robust threshold median + 4×MAD with an absolute floor of 0.5; 9×9 density window requiring 50% high-residual blocks; at least 12 blocks; mean residual ≥ 0.7; region/ring contrast ≥ 2.0.
Not available	Copy-move detection, Steganography checks — excluded from scoring in this build.

Verdict formula

```
weighted = sum(weight × module score) / sum(weight)
final = 0.55 × weighted + 0.45 × strongest module
```

Weights: file 0.15, metadata 0.20, ELA 0.20, copy-move 0.30, stego 0.15.

Band	Score	Meaning
Likely authentic	score below 20	No strong tampering signal; still requires examiner judgement.
Suspicious	score 20 to 49	At least one signal warrants review and corroboration.
Likely tampered	score 50 or above	A strong signal is consistent with manipulation.
Inconclusive	not scored	Not enough usable evidence to score.

Limitations

- ELA detects compression-history discontinuity, not intent. Smooth brightness changes and blurring can produce a low score even when the image was edited.
- A negative result is not proof of authenticity. Absence of evidence is not evidence of absence.
- Copy-move and steganography modules are not part of this build, so duplicated regions and LSB payloads are not assessed here.
- EXIF can be forged or stripped. Editor tags are leads, not findings of malicious intent.
- Thresholds are empirical and were not validated against a large labelled corpus.
- This tool is an educational prototype, not court-validated forensic software.

8. Declaration and chain of custody

The evidence file was opened read-only for this examination. All derived images, heatmaps and thumbnails were written under `output/`; the original bytes were never modified. The SHA-256 digest printed in Section 2 can be recomputed at any time to prove that the examined file is unchanged.

EXAMINED BY	Sagar
DATE (UTC)	2026-11-03 17:48:44
CASE ID	PP-20261103-174844-2509

SIGNATURE

Disclaimer

This report is the output of an automated screening tool. Findings are investigative leads, not legal proof. A qualified examiner must interpret them alongside the original evidence, the chain of custody and any corroborating sources. Scores are relative to the documented thresholds and do not express a probability of tampering.

Reproduction

```
python -m forensics.report samples/tampered/edited_metadata.jpg
python -m unittest discover -s tests -v
```

— End of report —